

# WASP-104b

R-2443

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-104\_180507 · created 2026-08-22T23:03:32+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458245.7799 +/- 0.0016 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.051 +/- 0.039                 |
| Transit depth (Rp/Rs)^2                           | 0.27 +/- 0.4 [%]                |
| Orbital Inclination (inc)                         | 88.54 +/- 3.0                   |
| Airmass coefficient 1 (a1)                        | 0.976 +/- 0.018                 |
| Airmass coefficient 2 (a2)                        | 0.0148 +/- 0.0041               |
| Scatter in the residuals of the lightcurve fit is | 1.84 %                          |
| Best Comparison Star                              | #2 - [117, 271]                 |
| Optimal Aperture                                  | 3.76                            |
| Optimal Annulus                                   | 9.29                            |
| Transit Duration (day)                            | 0.084 +/- 0.011                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.051 ± 0.039 vs 0.1214 (deviation 1.34σ) |
| Duration fit vs published | 0.084 d vs 0.0742 d (deviation 0.67σ)     |
| Mid-time O–C              | 4.1 min                                   |
| Depth SNR                 | 1.0                                       |
| Residual scatter          | 1.84 %                                    |

### Light curve

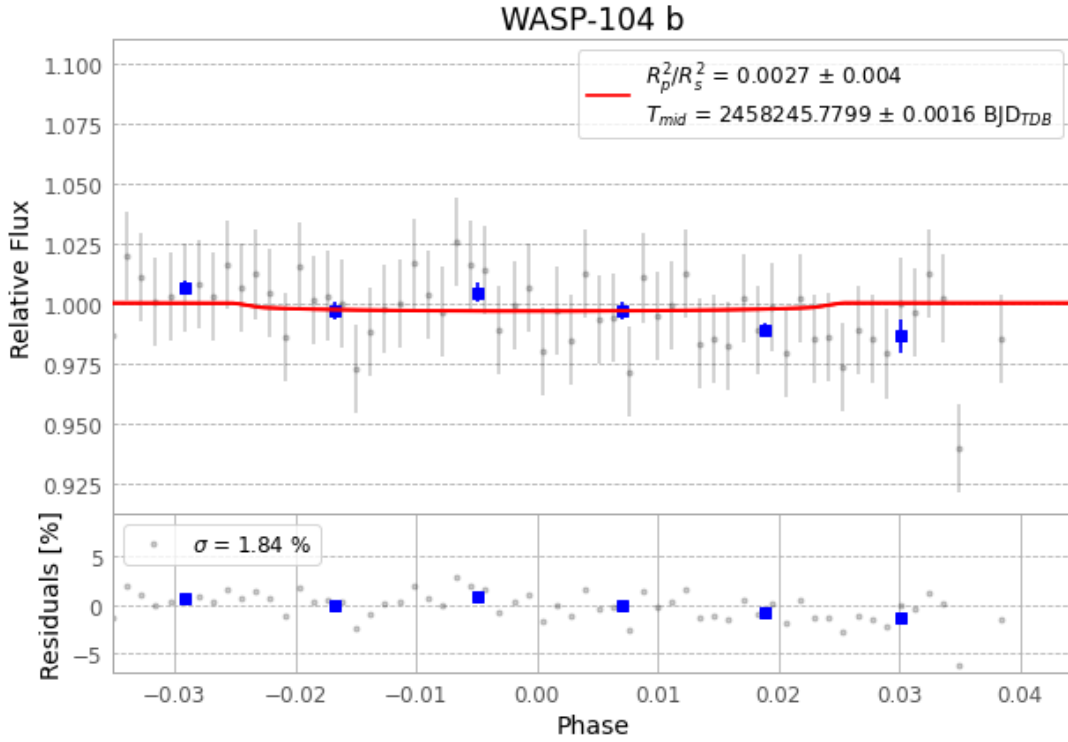


Figure 1 — FinalLightCurve\_WASP-104 b\_2018-05-06.png (SHA-256 7033d0a8d3db4372...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 261], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 a86fe51c875193ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

70 FITS frames (46 MB), WASP-104180507050915.FITS → WASP-104180507083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180507.log (f4b659a75c5e...), FinalLightCurve\_WASP-104 b\_2018-05-06.png (7033d0a8d3db...), FinalLightCurve\_WASP-104 b\_2018-05-06.csv (9016f96118d5...)

## Compute resources

Wall time 200.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:03Z.